SUPPLEMENTARY DIGITAL MATERIAL 2

Supplementary Table II.—Characteristics of the participants of the included studies. 34-54

Author and year	**Number of participants**	**Age of participants**	**Type of tumor**	**Inclusion critiria**	**Exclusion criteria**
Yu J *et al.* 2019 [34]	IG - 35 patients with a BT CG - 108 patients after a stroke	BT patients- 57.6±13.2 Stroke patients- 60.9±11.6	WHO grade (2016) I- 11 II-10 III-4 IV-10	BT or stroke defined as brain damage by computed tomography or magnetic resonance; admission to rehabilitation within <4 months of BT resection or stroke occurrence; age> 18 years; discharged for> 1 year	Combined neurodegenerative disease; and persistent minimally conscious state or vegetative status
Hojan K *et al.* 2020 [35]	Inpatient group- 28 patients Outpatient group- 26 patients	50.87±11.71	WHO grade (2016) I- 23 II-14 III-3 IV-13 I/II- 1	Stable medical course, age between 18 and 75, and good general condition, primary BT	Metastatics, significant co-morbidities or were medically unstable, had concomitant neurodegenerative disease, or severe psychiatric issues, or were in a persistent minimally conscious state or vegetative status as well as those BT patients who withdrew from the study before completing the 12 weeks of the

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
					rehabilitation program
Han E Y *et al.* 2015 [36]	BT patients- 29 Subacute stroke patients- 26	BT patients- 47.9±16.8 Stroke patients- 64.1±12.4	Benign: meningioma- 6, low-grade glioma- 4, other tumor types- 3 Malignant: metastatic lesions- 6, GBM- 5, other tumor types-5	BT (primary or metastatic), patients who could follow simple commands and complete the CNT	Patients unable to complete a questionnaire because of aphasia, visual and/or auditory problems, neglect, apraxia, or medical instability
Geler-Kulcu D *et al.* 2009 [37]	BT patients-21 Stroke patients- 21	BT group- 52.5±16 Stroke group- 56.7±11	Meningioma- 29% Glial tumor -71% GBM- 43%	All patients who were admitted to an inpatient rehabilitation program with a primary diagnosis of BT were invited to participate in the study. All patients with benign or malignant tumors after neurosurgery were included. Patients in a stable medical condition, tolerating one hour of therapy per day.	Radiotherapy or chemotherapy before rehabilitation program; neurological deficit after surgery with a need for more than one session of therapy
Greenberg E *et al.* 2006 [38]	BT patients- 168 Stroke patients - 1660	Meningioma group- 59.9±15.6 Glioma group- 54.1±13.4 Stroke group- 70.4±30.1	Intracranial meningiomas - 128 Cerebral gliomas- 40	Tumors of meningioma and glioma types were included. First-stroke patients (ischemic and hemorrhagic) were included.	Excluded were patients with recurring cerebral events and patients with repeat craniotomy.

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
Bilgin S *et al.* 2014 [39]	Patients with supratentorial BT- 40 Patients with supratentorial brain injury- 37	BT- 50.44±15.20 Brain injury- 44.23±17.03	Primary BT- 32 Metastatic BT- 8 Supratentorial High-grade gliomas with were selected	The age older than eighteen, radiological confirmation of BT diagnosis and brain injury with computed tomography scans, presence of a neurological deficit and clinical confirmation of medical and neurosurgical stabilities of patients to be admitted into rehabilitation program. In addition, none of the patients with BT received radiation therapy or chemotherapy through rehabilitation.	Excluded were patients who were diffuse axonal injury
Yoon J *et al.* 2015 [40]	IG- 20 CG- 20	IG- 48.6±11.3 CG- 50.0±17.5	Meningioma- 9 GBM- 10 Low-grade astrocytoma- 4 Metastatic tumor- 8 Schwannoma- 6 Others- 6	A BT diagnosis with a stable status after the completion of proper management (surgery, chemotherapy, or radiation therapy); patients with a Korean version Mini-Mental State Examination score of greater than 20 who are capable of understanding the therapist's	A diagnosis of global aphasia; decreased sitting balance; a recurrent BT; perceptual-cognitive dysfunction; musculoskeletal problems and/or peripheral neuropathy of the affected limb; and medical instability

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
				instructions; a Brunnstrom stage of higher than 2 corresponding to the affected upper extremity; a score on the Modified Ashworth spasticity scale of less than 2; and motor power grade of the affected shoulder of higher than poor.	
Huang M E *et al.* 2000 [41]	BT group-78 Brain injury group-78	BT group - 57.2±13.5 Brain injury group-58.6±15.3	benign and malignant	Medical stability, need for therapy from more than one discipline, demonstration of gains with acute care therapies, potential to tolerate 3 hour of therapy, willingness and motivation to participate in a rehabilitation program, and identifiable community supports and disposition to facilitate discharge.	Medical complications that prevented completion of rehabilitation or death during rehabilitation
Mukand J A *et al.* 2001 [42]	49	59.7	GBM- 16 Meningioma- 13 Metastasis- 13 Other- 2 Oligodendroglioma- 1	Adult patients admitted to the inpatient rehabilitation	-

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
Huang M E *et al.* 1998 [43]	BT group- 63 CVA group- 63	BT group 59.8±14.2 CVA group- 59.1±14.1	Benign and malignant	All persons admitted to inpatient rehabilitation with a primary diagnosis of BT.	Incomplete data set
Hansen A *et al.* 2018 [44]	24	62 (20-77)	WHO grade I- 1 II- 3 III- 2 IV- 18	Diagnosis of glioma as defined by the world health organization (WHO grade I–IV); age ≥18; KPS ≥70; and ability to follow Danish instructions	Pregnancy; active psychiatric diagnosis or substance abuse; heart problems exceeding New York Heart Association group III and IV; global aphasia.
Baima J *et al.* 2017 [45]	15	55±3.4	Astrocytoma -8 Metastasis- 7	Above 18 years of age, diagnosis of high-grade astrocytoma (grade 3 or 4), or any primary tumor with metastasis to the brain, at least 28 days out from surgery, currently undergoing follow-up for standard oncology care at our center, and able to perform exercises independently at home.	Untreated thromboembolic disease; on home oxygen
Roberts P S *et*	Rehabilitation group- 100	Rehabilitation group- 61.7±14	GBM	Diagnose GBM between 2003 and 2010, with survival	Patients were excluded because they

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
al. 2014 [46]	Non rehabilitation group- 312	Non rehabilitation group- 59.9±13.7		data updated through January 23, 2013	died within 30 days of surgery or their surgery dates were not available
Reilly JM *et al.* 2020 [47]	Initial diagnosis of GBM - 25 Recurrent GBM - 25	initial GBM- 61.6 recurrent GBM- 60.8	GBM	Any patient over the age of 18 years who was admitted to the acute inpatient rehabilitation with a diagnosis identified by the International Classification of Diseases, Tenth Revision (ICD-10) codes C71 (malignant neoplasm of the brain), D33.2 (glioma), and C71.9 (GBM), D49.6 (neoplasm of unspecified behavior), and D44.4 (neoplasm of uncertain behavior).	Patients were excluded from the recurrent GBM group if their neuroimaging did not reveal progression of disease or recurrence.
Marciniak CM *et al.* 2001 [48]	132	58±16.7	Metastasis- 21 Astrocytoma- 34 Meningioma -44 Other- 33	Patients diagnosed with a primary or secondary BT who were at or over the age of 18 years and were discharged from our comprehensive, free-standing, university-affiliated, inpatient acute rehabilitation program within a	no complete FIM data

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
				3-year period (Sept 1993-Aug 1996)	
O'Dell M W *et al.* 1998 [49]	40	BT group- 53.1±15.4	GBM- 20% Meningioma- 20% Astrocytoma- 12,5% Metastatic- 12.5% Oligodendroglioma- 5% Pituitary adenoma 5% Other- 25%	All patients with a diagnosis of BT admitted to The Drake Center Brain Injury Rehabilitation Unit from March 1994 to December 1996.	Neurological deficits caused by an unrelated or history of recurrent meningioma admitted for deconditioning after multiple seizures without a documented new cerebral lesion.
Ayotte S L *et al.* 2017 [50]	16	53.75±14.7	GBM- 5 Meningioma- 4 Metastatic tumor- 4 Astrocytoma- 1 Other- 2	Age of 18 to 80 years, diagnosis of a primary or metastatic BT, hemoglobin level greater than 7.5 and platelets greater than 30 000, and tolerance to sit upright and perform lower-body ergometer training for more than 5 minutes	A history of unrelated progressive neurologic disorders, orthopedic conditions preventing participation in ergometer training, preexisting bone metastases at risk for fracture, and any unstable cardiopulmonary issues or hospitalizations in the past 6 months.

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
Shahpar S *et al.* 2018 [51]	46	54 (18-92)	GBM – 26 Astrocytomas- 8 Oligoastrocytoma-1 Oligodendrogliomas- 5 Metastases- 4 Lymphoma- 2	Age 18 or older with a diagnosis of a malignant BT and participating in the day rehabilitation program.	Diagnosis of benign or low grade BT and tumors that are more often seen in the pediatric population, including medulloblastoma, craniopharyngioma, and Rathke's cleft cyst. Other patients excluded include those with a prior history of neurological disease, such as brain injury or stroke that could lead to impairments and functional deficits unrelated to the BT and/or cancer treatment.
Bartolo M *et al.* 2012 [52]	BT group- 75 Stroke group- 75	BT group- 63.5±13.2 Stroke group- 65.7±10.6	GBM- 43 Meningioma- 32	Patients were admitted within two weeks of surgery. No adjuvant therapy, nor radiotherapy or chemotherapy before or during the rehabilitation program.	Patients with oligoastrocytoma, oligodendroglyoma, and ependimomas were excluded from the study

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
Fu J B *et al.* 2019 [53]	21	<65 15 (50%) ≥65 15 (50%)	LMD	Acute inpatient rehabilitation admissions data from 8 January 2014 to 31 August 2018 at a tertiary referral based National Cancer Institute (NCI) Designated Comprehensive Cancer Center of patients with a pathology confirmed diagnosis of LMD within 6 months of rehabilitation transfer	Excluded patients who did not have a CSF pathology confirmed record of LMD within 6 months of acute inpatient rehabilitation admission
Khan F *et al.* 2014 [54]	IG- 53 Waitlist CG - 53	51±13.6	Glioma WHO grade I- 14 II- 30 III- 15 IV- 37	Aged ≥18 years and fulfilled standard diagnostic criteria for BT grading system as outlined by the WHO for central nervous system tumours; had completed definitive treatment (and not currently on treatment) and assessed by a surgeon/oncologist at the RMH. Stable medical course, post BT surgery, radiotherapy and/or chemotherapy, assessed by a rehabilitation physician/neurosurgeon for presence	Those who had benign or metastatic BTs, significant co-morbidities or medically unstable, or psychiatric disorders (such as uncontrolled schizophrenia, actively suicidal/self-harm or physically aggressive (based on clinical judgement)) limiting participation in rehabilitation, those bed-bound

Author and year	Number of participants	Age of participants	Type of tumor	Inclusion critiria	Exclusion criteria
				of neurological deficits and ability to participate in therapy up to 2.5 h of interrupted therapy/day; and the clinical judgment of the assessing rehabilitation team participants resided in the community (area of greater Melbourne<60 km radius), and were able to communicate in English.	and/or institutionalized in nursing homes were excluded

BT: brain tumor; WHO: World Health Organization; IG: intervention group; CG: control group; CVA: cerebrovascular accident; GBM: glioblastoma multiforme; LMD: leptomeningeal disease.

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